

Calculus II Lesson 20

Root Test for Infinite Series $\sum_{n=1}^{\infty} a_n$

a) If $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} < 1$, then the series $\sum_{n=1}^{\infty} a_n$ converges.

b) If $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} > 1$, then the series $\sum_{n=1}^{\infty} a_n$ diverges.

c) If $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = 1$, use another test.

1) $\sum_{n=1}^{\infty} \frac{1}{5^n}$ is convergent or divergent?

Note: $a_n = \frac{1}{5^n}$

a) $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \lim_{n \rightarrow \infty} \sqrt[n]{\left|\frac{1}{5^n}\right|} = \underline{\hspace{2cm}} ?$

b) What does the Root Test say about the convergent/divergent of the series $\sum_{n=1}^{\infty} \frac{1}{5^n}$?

2) $\sum_{n=1}^{\infty} \left(\frac{n}{2n+1} \right)^n$ is convergent or divergent?

Note: $a_n = \left(\frac{n}{2n+1} \right)^n$

a) $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{n}{2n+1} \right)^n} = \underline{\hspace{2cm}}$ Hint: Use L'Hopital's Rule

- A) 1 B) 2 C) $\frac{1}{2}$

b) What does the Root Test say about the convergent/divergent of the series $\sum_{n=1}^{\infty} \left(\frac{n}{2n+1} \right)^n$?

3) $\sum_{n=1}^{\infty} \left(\frac{5n+2}{n+3} \right)^n$ is convergent or divergent?

Note: $a_n = \left(\frac{5n+2}{n+3} \right)^n$

a) $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \left(\frac{5n+2}{n+3} \right)^n \right|} = \underline{\hspace{2cm}}$ Hint: Use L'Hopital's Rule

A) $\frac{2}{3}$

B) 5

C) $\frac{1}{5}$

b) What does the Root Test say about the convergent/divergent of the series $\sum_{n=1}^{\infty} \left(\frac{5n+2}{n+3} \right)^n$?

4) $\sum_{n=1}^{\infty} (2\sqrt[n]{n} + 1)^n$ is convergent or divergent?

Note: $a_n = (2\sqrt[n]{n} + 1)^n$

a) $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \lim_{n \rightarrow \infty} \sqrt[n]{(2\sqrt[n]{n} + 1)^n} = \underline{\hspace{2cm}}$ Hint: $\lim_{n \rightarrow \infty} \sqrt[n]{n} = \lim_{n \rightarrow \infty} n^{1/n} = 1$.

A) 1

B) 2

C) 3

b) What does the Root Test say about the convergent/divergent of the series $\sum_{n=1}^{\infty} (2\sqrt[n]{n} + 1)^n$?

5) $\sum_{n=1}^{\infty} \frac{n}{3^n}$ is convergent or divergent?

Note: $a_n = \frac{n}{3^n}$

a) $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \lim_{n \rightarrow \infty} \sqrt[n]{\left| \frac{n}{3^n} \right|} = \underline{\hspace{2cm}}$ Hint: $\lim_{n \rightarrow \infty} \sqrt[n]{n} = \lim_{n \rightarrow \infty} n^{1/n} = 1$.

A) 1 B) $\frac{1}{3}$ C) 3

b) What does the Root Test say about the convergent/divergent of the series $\sum_{n=1}^{\infty} \frac{n}{3^n}$?

